

Symbols and Syntax

of Predicate Logic

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- 1 Symbols and Syntax
- 2 Translation

Predicate

$predicate(p_1, p_2, \dots, p_n)$

- ① *predicate*:
 - a valid identifier (the first character: lower case).
 - evaluated to either **true** or **false** when all variables are quantified or binded to constants
- ② $p_1, p_2, \dots, p_n$: parameter, can be one of the following
 - Variables (the first character: lower case).
 - Constants
 - Functions. For examples, $f(x)$, $f(x, y)$, etc.

Predicate

$\text{predicate}(p_1, p_2, \dots, p_n)$

To represent compound statements

- Use five operators:
 - $\neg$: negation
 - $\wedge$: conjunction
 - $\vee$: disjunction
 - $\Rightarrow$: implication
 - $\Leftrightarrow$: equivalence
- Use quantifiers:
 - $\forall x$: for all x
 - $\exists x$: exists x

Three cases of statements in predicate logic

① Singular statement

- is a statement that makes an assertion about a **specifically named** person, place, thing, or time.

② Universal statement

- is a statement that makes an assertion about **every member** of its subject class.

③ Particular statement

- is a statement that makes an assertion about one or more **un-named members** of the subject class.

Translation

Singular statement

English	Predicates	Propositions
Socrates is mortal.	<i>mortal</i> (" Socrates")	<i>S</i>
Aristotle is mortal.	<i>mortal</i> (" Aristotle")	<i>A</i>
Hanoi is the capital of Vietnam.	<i>capital</i> (" Hanoi")	<i>H</i>
DaNang is not the capital of Vietnam.	$\neg$ <i>capital</i> (" DaNang")	<i>D</i>
Huong is a student.	<i>student</i> (" Huong")	<i>Huong</i>
Tam is a student.	<i>student</i> (" Tam")	<i>Tam</i>

Universal statement

English	Predicates	Meaning
All S are P.	$\forall x(s(x) \Rightarrow p(x))$	For any x, if x is an S, then x is a P.
No S are P.	$\forall x(s(x) \Rightarrow \neg p(x))$	For any x, if x is an S, then x is not

Universal statement

English	Predicates
All skyscrapers are tall.	$\forall x(skyscraper(x) \Rightarrow tall(x))$
No frogs are birds.	$\forall x(frog(x) \Rightarrow \neg bird(x))$
All ambassadors are statesmen.	$\forall x(ambassador(x) \Rightarrow statesman(x))$
No diamonds are rubies.	$\forall x(diamond(x) \Rightarrow \neg ruby(x))$

Particular statement

English	Predicates
All skyscrapers are tall.	$\forall x(skyscraper(x) \Rightarrow tall(x))$
No frogs are birds.	$\forall x(frog(x) \Rightarrow \neg bird(x))$
All ambassadors are statesmen.	$\forall x(ambassador(x) \Rightarrow statesman(x))$
No diamonds are rubies.	$\forall x(diamond(x) \Rightarrow \neg ruby(x))$

Particular statement

English	Predicates	Meaning
Some S are P.	$\exists x(s(x) \wedge p(x))$	There exists an x such that x is an S and x is a P.
Some S are not P.	$\exists x(s(x) \wedge \neg p(x))$	There exists an x such that x is an S and x is not a P.

Particular statement

English	Predicates
Some students are businessmen.	$\exists x(student(x) \wedge businessman(x))$
Some businessmen are students.	$\exists x(businessman(x) \wedge student(x))$
Some vehicles are not motorcycles.	$\exists x(vehicle(x) \wedge \neg motorcycle(x))$
Some jobs are boring	$\exists x(job(x) \wedge boring(x))$
Some jobs are not boring	$\exists x(job(x) \wedge \neg boring(x))$
Some diseases are not contagious.	$\exists x(disease(x) \wedge contagious(x))$

Translation Guidelines

- ❶ Important!: Understand the meaning of the statement to be translated.
- ❷ If the statement makes an assertion about every member of its subject class, a universal quantifier should be used to translate it.
- ❸ If it makes an assertion about only one or more members of this class, an existential quantifier should be used.

Translation

Examples

English	Predicates
There are happy marriages.	$\exists x(\text{marriage}(x) \wedge \text{happy}(x))$
Every pediatrician loses sleep.	$\forall x(\text{ped}(x) \Rightarrow \text{losesleep}(x))$
Animals exist.	$\exists x(\text{animal}(x))$
Unicorns do not exist.	$\neg \exists (\text{unicorn}(x))$
Anything is conceivable	$\forall x(\text{conceivable}(x))$
Sea lions are mammals.	$\forall x(\text{sealion}(x) \Rightarrow \text{mammal}(x))$
Sea lions live in these caves.	$\exists x(\text{sealion}(x) \wedge \text{livecave}(x))$
Egomaniacs are not pleasant companions.	$\forall x(\text{egomaniac}(x) \wedge \neg \text{pleasant}(x))$

Translation

Examples

English	Predicates
A few egomaniacs did not arrive on time.	$\exists x(egomaniac(x) \wedge \neg arrive_o t(x))$
Only close friends were invited to the wedding. (invited $\Rightarrow$ close friend)	$\forall x(invited(x) \Rightarrow closefriend(x))$
None but citizens are eligible to vote.	$\forall x(eligible(x) \Rightarrow citizen(x))$
All eligible people are citizens	$\neg(\forall x(girl(x) \Rightarrow sell(x)))$
It is not the case that every Girl Scout sells cookies.	$\exists x(girl(x) \wedge \neg sell(x))$
Not a single psychologist attended the convention.	$\neg[\exists x(psychologist(x) \wedge attended(x))]$
	$\forall x(psychologist(x) \Rightarrow \neg attended(x))$

Translation

Complex Translation

English	Predicates
1.	Only snakes and lizards thrive in the desert. $\forall x(thrive(x) \Rightarrow (snake(x) \vee lizard(x)))$
2.	Oranges and lemons are citrus fruits. $\forall x((oranges(x) \vee lemon(x)) \Rightarrow citrus(x))$
3.	Cats and dogs bite if they are frightened or harassed.. $\forall x\{(cat(x) \vee dog(x)) \Rightarrow [(fright(x) \vee harassed(x)) \Rightarrow bite(x)]\}$
4.	Ripe apples are crunchy and delicious. $\forall x[(ripe(x) \wedge apple(x)) \Rightarrow (crunchy(x) \wedge deli(x))]$
5.	Peaches are edible unless they are rotten. $\forall x[peach(x) \Rightarrow (\neg rotten(x) \Rightarrow edible(x))]$ $\forall x[peach(x) \Rightarrow (edible(x) \vee rotten(x))]$